

Robins K Roy

Date of Birth: 16 November 2002

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Professional Summary

Aspiring AI Engineer with hands-on experience in Machine Learning, NLP, and Generative AI. Selected for Google Cloud's GenAI Exchange Program. Certified by IBM, NVIDIA, and Infosys. Experienced in predictive modeling, AI systems, and real-world deployment with strong leadership and research skills.

Education

Master of Computer Applications (MCA) 2024 – Present
Marian College Kuttikkanam (Autonomous), Mahatma Gandhi University

Bachelor of Computer Applications (BCA) 2021 – 2024
MES College Nedumkandam, Mahatma Gandhi University

Certifications

- Artificial Intelligence GenAI – NVIDIA (2025)
- Machine Learning – IBM (2025)
- Data Analysis with Python – IBM (2024)
- Introduction to Data Science – Infosys (2025)
- Introduction to NLP – Infosys (2025)
- Gen AI Intensive – Kaggle (2025)
- Data Science Methodology – IBM (2024)
- Data Science Tools – IBM (2024)
- Prompt Engineering for Everyone – IBM (2024)
- Data Science and ML Workshop – Hajee KRH College (2022)

Projects

- Autograd Engine:** Built a custom Python-based automatic differentiation engine inspired by Karpathy's micrograd, enabling gradient computation through reverse-mode autodiff for neural networks.
- AI Tutor Mascot:** Developed an intelligent AI tutor using FastAPI, LangChain, ChromaDB, and OpenAI APIs. Integrated speech recognition, emotion-aware response generation, and real-time chat with local LLM support using Ollama.
- AttendAI – AI-based Attendance Automation System:** Designed a Flask and TensorFlow-powered real-time attendance system using MTCNN and FaceNet for face detection and recognition. Achieved 99% accuracy with cosine similarity-based matching and SQLite database for attendance reports.
- Vehicle CO₂ Estimator:** Used Multiple Linear Regression to model vehicle emissions from engine parameters.
- Iris Flower Classification:** Leveraged K-Nearest Neighbors (KNN) algorithm to classify Iris species using petal and sepal dimensions, optimizing parameters to reduce misclassification.

Technical Skills

Languages: Python, C++, SQL, Java, HTML, CSS, Linux

Libraries/Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, OpenCV, MTCNN, FaceNet, Flask, FastAPI, LangChain, Ollama, Django, Streamlit

Tools Platforms: Vertex AI, Google Cloud, Kaggle, GitHub, Jupyter, Power BI, Excel, MongoDB, Hugging Face, AWS, Render

Key Concepts: Deep Learning, Generative AI, NLP, RAG Architecture, Local LLM Integration, Prompt Engineering, Regression, Classification, Model Evaluation, Feature Engineering

Internships Experience

AI Intern – IBM Jun 2025 – Aug 2025

- Completed hands-on AI/ML projects using Python, Scikit-learn, and TensorFlow.
- Gained practical skills in NLP, model evaluation, and generative AI applications.

AI Engineer – Tabernacle Pvt Ltd, Singapore Jul 2025 – Sep 2025

- Developed multiple AI-driven applications, including voice-to-voice APIs and automation workflows.
- Designed and deployed Python automation projects to streamline data processing tasks.
- Collaborated on building scalable AI models for real-world use cases.

GenAI Exchange Program Participant – Google Cloud May 2025 – Present

- Trained in Vertex AI, Gemini, Imagen, and prompt design.
- Gaining real-world experience in deploying AI apps with Gemini API, RAG, and Streamlit.

Research Conferences

International Conference (2025):

Presented “Advancement in Agricultural Robotics and Soil Nutrient Analysis” at Rajagiri College of Management and Applied Sciences.

National Conference (2024):
Presented “AI for Sustainability: Balancing Innovation with Ethical Responsibility” at Marian College Kuttikkanam.

Achievements

UNAI Student Coordinator – MES College Nedumkandam	2022 – 2024
Led national-level initiatives under the UN Academic Impact banner; promoted sustainable development and education access.	
Workshop Speaker – MES College Nedumkandam	2025
Conducted a workshop on “Leveraging AI Tools and LinkedIn for Career Growth” for undergraduate students.	